MILITARY SPECIFICATION

SWITCHES, DUAL IN-LINE PACKAGE (DIP),
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-S-83504A, dated 31 December 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

Following 3.3.4, add new paragraph as follows:

"3.3.5 Tin plated finishes. Use of pure tin plating is prohibited (see 6.8). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

PAGE 4

3.4.7, last sentence, delete and substitute:

"Unless otherwise specified (see 3.1), there shall be no meniscus on the switch leads from the terminal potting/sealing material within 0.010 inch the end of the solder standoffs.

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3.5.15, last sentence, delete and substitute:

"Unless otherwise specified (see 3.1), there shall be a minimum 0.010 inch clearance between the end of the solder standoff and any sealing potting/epoxy on the pins."

PAGE 8

TABLE II, Group V, column four, delete "16 samples" and substitute "8 samples".

TABLE II, delete 2/ at the end of the table and substitute:

"2/ Two samples for each of two cleaning solutions in immersion, and an equal number for spray cleaning.

TABLE II, Group V, column one, following "2/", add "4/". Also, following 3/ at the end of the table, add 4/ as follows:

"4/ Group I tests are not required for group V samples."

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TABLE IV, delete and substitute:

"TABLE IV. Zero defect sampling plan.

Lot size	Number of switches to be tested		
	Major 1/	Minor 1/	Cosmetic 1/
1-8	All	5	3
9-15	13	5	3
16-25	13	5	3
26-50	13	5	5
51-90	13	7	6
91-150	13	11	7
151-280	20	13	10
281-500	29	16	11
501-1,200	34	19	15
1,201-3,200	42	23	18
3,201-10,000	50	29	22
10,001-35,000	60	35	29

1/ Major, minor, and cosmetic defects are defined in section 6. Inspection for minor and cosmetic defects is applicable to visual and mechanical examination only."

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4.8.3, following the first sentence, add "Measurement of the operating force of the switches with slide actuation shall be made with the measuring device operated parallel to the normal actuation plane of the slide."

PAGE 15

4.8.17, delete subparagraph c.

4.8.18, delete subparagraphs d, d(1), and d(2).

4.8.19, delete and substitute:

"4.8.19 Machine soldering (see 3.5.18). Switches shall be tested in accordance with EIA RS-448-2, method 16. Fluxing method A (touching body) shall be used; solder time shall be 4 seconds; and cleaning shall be accomplished by immersion and spray, using isopropyl alcohol and deionized water/detergent only. Switches with tape top seal shall be tested with seal in place."

PAGE 17

Following 6.7, add new paragraph as follows:

"6.8 Tin plated finishes. Use of tin plating is prohibited (see 3.3.5). Use of tin finishes can result in the tin whisker growth. Tin whisker growth can result in adverse effects on the operation of electronic equipment systems. For additional information on this matter, refer to ASTM B545-83 (Standard Specification for Electrodeposited Coating of Tin)."

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AV
Navy - AS
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5930-1507)

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